

How do we use a test point e.g. (1,0) to sketch phase portraits of spiral, center and degenerate nodes?

Answer:

To sketch phase portraits for spiral points, centers, and degenerate nodes using a test point like (1, 0), you need to determine the direction of the vector field at that point. This helps in understanding the local behavior around the equilibrium point. Here's how to proceed for each type:

1. Spiral Point:

- The trajectories around the origin are spirals.
- To determine the direction of spiraling, compute the derivative $\mathbf{y}'$ at the test point (1, 0)
- Use the matrix $\mathbf{A}$ from the system $\mathbf{y}' = \mathbf{A}\mathbf{y}$. The first column of $\mathbf{A}$ given by $\begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}$ represents $\mathbf{y}'$ at (1, 0).
- The sign of a_{21} determines the rotational direction: if $a_{21} > 0$, it's counterclockwise; if $a_{21} < 0$, it's clockwise.

2. Center:

- Trajectories form concentric closed loops.
- Similar to a spiral point, compute $\mathbf{y}'$ at (1, 0).
- Similar to the spiral point case, the sign of a_{21} tells you whether the motion is CCW or CW.

3. Degenerate Node:

- Characterised by a repeated eigenvalue and a single (independent) eigenvector.
- Compute $\mathbf{y}'$ at (1,0), which is again given by $\begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}$.
- If $\lambda < 0$, trajectories move towards the origin.
 - If $a_{21} > 0$, the trajectories start far away from origin below the horizontal axis. They cross the horizontal axis before making a turn and parallel to the eigenvector line as they move towards the origin.
 - If $a_{21} < 0$, the trajectories start far away from origin above the horizontal axis. They cross the horizontal axis before making a turn and parallel to the eigenvector line as they move towards the origin.
- If $\lambda > 0$, trajectories move away from the origin.

- If $a_{21} > 0$, the trajectories start near the origin below the horizontal axis. They cross the horizontal axis before making a turn and parallel to the eigenvector line as they move away from the origin.
- If $a_{21} < 0$, the trajectories start near the origin above the horizontal axis. They cross the horizontal axis before making a turn and parallel to the eigenvector line as they move away from the origin.

Using a test point helps you confirm the direction and shape of trajectories by evaluating the local arrow direction given by $\mathbf{y}' = \mathbf{A} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. This insight is essential for correctly drawing the phase portrait according to the type of equilibrium point.